Introduction

The purpose of this document is to evaluate potential security weaknesses and implement security controls within the mobile app's infrastructure.

Scope of Work

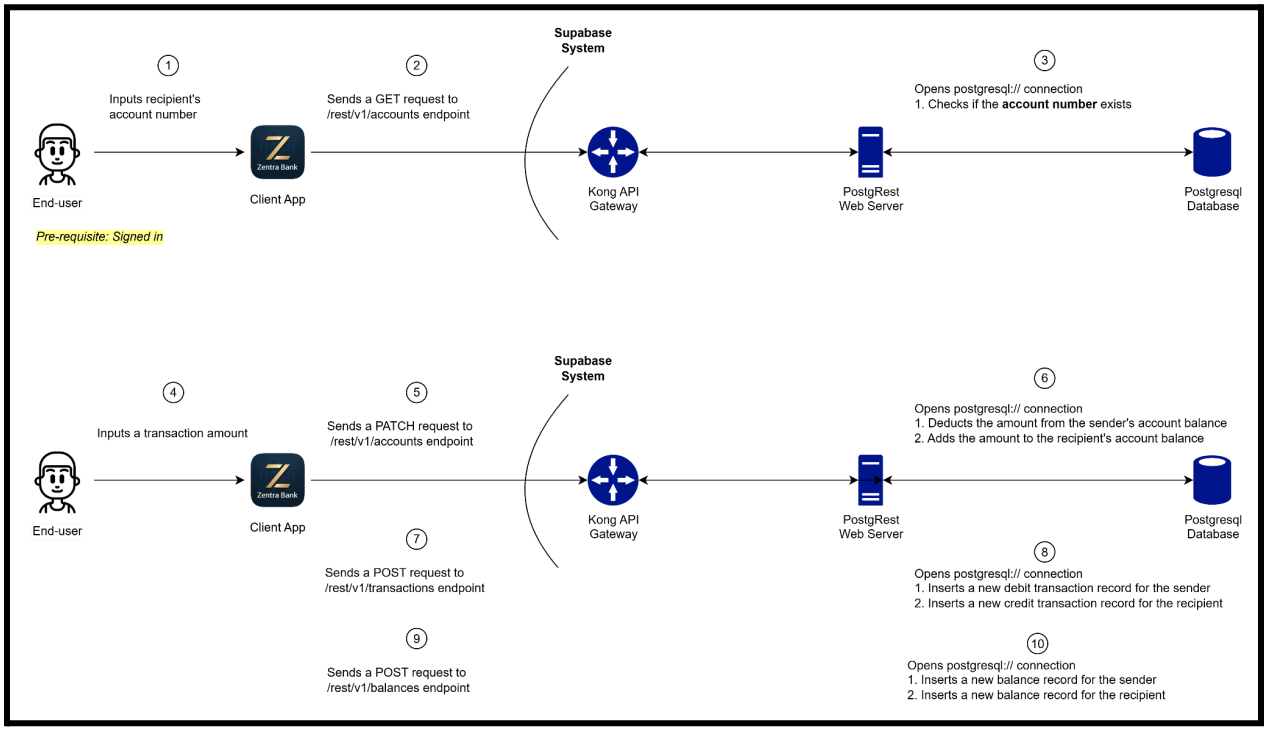
This exercise focuses on 1 user story within the Zentra Bank mobile app:

1. As a user, I want to **transfer funds to other accounts via account number** so that I can send money to friends, family, or service providers easily

Out of Scope

This exercise strictly focuses on the mobile application and its functionalities. Any components outside the mobile app, which includes but not limited to REST APIs, server infrastructure, remote database, and external services, are not within the scope.

User story 4



The data flow for this user story is as follows:

"As a user, I want to transfer funds to other accounts via account number so that I can send money to friends, family, or service providers easily."

1. The End-user inputs the recipients' account number
2. The Client App (CA) sends an account number check request via REST API to the Web Server (WS) to verify whether it exists
3. The Web Server (WS) checks 1) if the account number exists against the `accounts` table in the Database (DB) via Postgres connection

```
GET /rest/v1/accounts?select=*&account_number=eq.00177714179 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
accept: application/json
accept-profile: public
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9
2MjQsImV4cCI6MjA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc
2F1dGgvdjEiLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhd
TY4OTIyMzIsImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZ
l19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92Z
mMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0Z
nRpbWVzdGFtcCI6MTc1Njg5MjIzMn1dLCJzZXNzaW9uX2lkIjoiNGYzYmJiOWMtNzE0OC00MGEyL
4_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
x-client-info: supabase-js-react-native/2.49.5-next.1
```

4. The End-user inputs a transaction amount

5. To complete the transfer, the Client App (CA) first sends a request to update accounts' balances via REST API to the Web Server (WS) to 1) deduct the amount from the sender's account balance and 2) add the amount to the recipient's account balance

6. The Web Server (WS) makes an update query to the **`accounts`** table in the Database (DB) via Postgres connection. The Database (DB) then 1) deducts the amount from the sender's account balance and 2) adds the amount to the recipient's account balance

```
PATCH /rest/v1/accounts?account_id=eq.4fc4ca5e-09d2-433c-9321-add4c91aa6a2 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cH
2MjQsImV4cCI6MjA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRw
2F1dGgvdjEiLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhdWQiOiJhdXRo
TY4OTIyMzIsImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZGF0YSI6eyJw
l19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92ZXJpZmllZCI6
mMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0ZWQiLCJhYWwi
nRpbWVzdGFtcCI6MTc1Njg5MjIzMn1dLCJzZXNzaW9uX2lkIjoiNGYzYmJiOWMtNzE0OC00MGEyLTljNTctMzc1
4_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 21

{"current_balance":2}
```

```
PATCH /rest/v1/accounts?account_id=eq.15872f95-e28a-4dc6-9bdc-6ffc22e6e0b2 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cHN
2MjQsImV4cCI6MjA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRw
2FldGgvdjEiLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhdWQiOiJhdXRo
TY4OTIyMzIsImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZGF0YSI6eyJw
l19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92ZXJpZmllZCI6
mMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0ZWQiLCJhYWwi
nRpbWVzdGFtcCI6MTc1Njg5MjIzMn1dLCJzZXNzaW9uX2lkIjoiNGYzYmJiOWMtNzE0OC00MGEyLTljNTctMzc1
4_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 23

{"current_balance":18}
```


7. The Client App (CA) will also send a request to insert new transaction records via REST API to the Web Server (WS) for audit trail

8. The Web Server (WS) makes an insert query to the **`transactions`** table in the Database (DB) via Postgres connection. The Database (DB) then inserts 1) a debit transaction record for the sender and 2) a credit transaction record for the recipient


```
POST /rest/v1/transactions?columns="account_id","booking_datetime","amount","currency","credit_debit","description","transaction_type","status" HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cHNwYXVwYmFoIiwicm9sZSI6ImFub24iLCJpYXQiOjE3NDg0ODA2MjQsImV4cCI6MjA2NDA1N
BmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRwczovL3ZvbW9oY2NtYXB2cHNwYXVwYmFoLnN1cGFiYXNlLmNvL2F1dGgvdjEiLCJzdWIiOiI
C00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0ZWQiLCJleHAiOjE3NTY4OTU4MzIsImlhdCI6MTc1Njg5MjIzMiwidXNlcl9tZXRhZGF0YSI6eyJlbWFpbCI6InRlc3RuaWNAbG1haWwuY29tIn1
tcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 430

[{
  "account_id": "4fc4ca5e-09d2-433c-9321-add4c91aa6a2",
  "booking_datetime": "2025-09-03T09:49:48.381Z",
  "amount": 1,
  "currency": "SGD",
  "credit_debit": "Debit",
  "description": "Transfer to 00177714179",
  "transaction_type": "Online Transfer",
  "status": "Booked"
},{
  "account_id": "15872f95-e28a-4dc6-9bdc-6ffc22e6e0b2",
  "booking_datetime": "2025-09-03T09:49:48.381Z",
  "amount": 1,
  "currency": "SGD",
  "credit_debit": "Credit",
  "description": "Transfer from 00139540222",
  "transaction_type": "Online Transfer",
  "status": "Booked"
}]
```


9. The Client App (CA) will also send a request to insert new balance records via REST API to the Web Server (WS) for audit trail

10. The Web Server (WS) makes an insert query to the **`balances`** table in the Database (DB) via Postgres connection. The Database (DB) then inserts a new balance record for 1) the sender and 2) the recipient

```
POST /rest/v1/balances?columns="account_id","balance_type","amount","currency","credit_debit","reference_date" HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cHNwYXVwYmFoIiwicm9sZSI6ImFub24iLCJpY
jA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRwczovL3ZvbW9oY2NtYXB2cHNwYXVwYmFoLnN
zdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImFldGhlbnRpY2F0ZWQiLCJleHAiOjE3NTY4OTU4MzIsImlhdCI6M
GF0YSI6eyJlbWFpbCI6InRlc3RuaWNAbGlhaWwuY29tIn19.XmKw04_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 373

[{
  "account_id": "4fc4ca5e-09d2-433c-9321-add4c91aa6a2",
  "balance_type": "InterimAvailable",
  "amount": 2,
  "currency": "SGD",
  "credit_debit": "Debit",
  "reference_date": "2025-09-03T09:49:48.514Z"
},{
  "account_id": "15872f95-e28a-4dc6-9bdc-6ffc22e6e0b2",
  "balance_type": "InterimAvailable",
  "amount": 18,
  "currency": "SGD",
  "credit_debit": "Credit",
  "reference_date": "2025-09-03T09:49:48.514Z"
}]
```